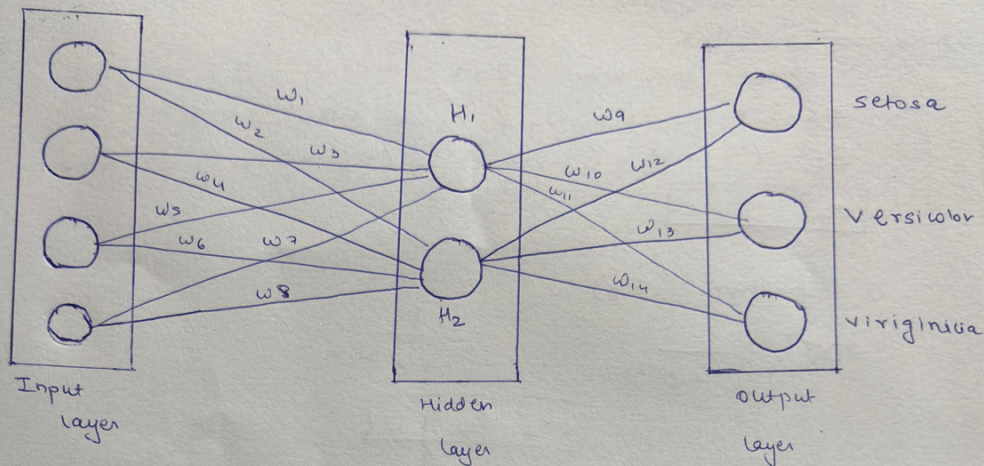


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DL (Mathematical Calculation single - Hidden-layer Neural Network)



1.] Given Parameters.

- Input Vector : $X = [5.1, 3.2, 1.4, 0.2]$
- Hidden layer weight (H_1) : $w_1 = 1$ $w_3 = 3$
 $w_5 = 5$ $w_7 = 7$
- Hidden weights (H_2) : $w_2 = 2$ $w_4 = 4$
 $w_6 = 6$ $w_8 = 8$
- Output weights (from H_1) : $w_9 = 1.3$, $w_{10} = 0.5$, $w_{11} = 1.2$
- Output weight (from H_2) : $w_{12} = 0.5$, $w_{13} = 0.2$, $w_{14} = 0.4$
- Biases : $b_1 = 2$ $b_2 = 0.1$

2.] Hidden Layer Calculation

A) Neuron 1 (H_1):

• Linear combination (Z_1):

$$Z_1 = x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4 + b_1$$

$$Z_1 = (5.1 \times 1) + (3.2 \times 2) + (1.4 \times 3) + (0.2 \times 5) + 2$$

$$Z_1 = 5.1 + 6.4 + 4.2 + 1.0 + 2$$

$$\boxed{Z_1 = 28.7}$$

• Activation (σ_1):

$$\sigma_1 = \frac{1}{1 + e^{-z_1}} = \frac{1}{1 + e^{-19.1}}$$

$$\sigma_1 \approx 1.0$$

B) Neuron 2 (H_2):

• Linear Combination (z_2):

$$z_2 = x_1 w_2 + x_2 w_4 + x_3 w_6 + x_4 w_8 + b_1$$

$$z_2 = (5.1 \times 2) + (3.2 \times 4) + (1.6 \times 6) + (0.2 \times 8) + 2$$

$$z_2 = 35.0$$

• Activation (σ_2):

$$\sigma_2 = \frac{1}{1 + e^{-z_2}} = \frac{1}{1 + e^{-35.0}}$$

$$\sigma_2 \approx 1.0$$

3.) Output Layer Combination Calculation [Combining both neurons]

A) Setosa (z_{setosa}):

$$\begin{aligned} z_{\text{setosa}} &= (\sigma_1 \times w_{1a}) + (\sigma_2 \times w_{12}) + b_2 \\ &= (1.0 \times 1.3) + (1.0 \times 0.5) + 0.1 \\ &= 1.3 + 0.5 + 0.1 \end{aligned}$$

$$z_{\text{setosa}} = 1.9$$

B) Versicolor ($z_{\text{versicolor}}$):

$$\begin{aligned} z_{\text{versicolor}} &= (\sigma_1 \times w_{1b}) + (\sigma_2 \times w_{13}) + b_2 \\ &= (1.0 \times 0.9) + (1.0 \times 0.2) + 0.1 \\ &= 0.9 + 0.2 + 0.1 \end{aligned}$$

$$z_{\text{versicolor}} = 0.8$$

C) Virginica ($z_{\text{virginica}}$):

$$\begin{aligned} z_{\text{virginica}} &= (\sigma_1 \times w_{1c}) + (\sigma_2 \times w_{14}) + b_2 \\ &= (1.0 \times 1.2) + (1.0 \times 0.4) + 0.1 \\ &= 1.2 + 0.4 + 0.1 \end{aligned}$$

$$z_{\text{virginica}} = 1.7$$

nal Prediction

•) Output Vector : $[1.9, 0.8, 1.7]$

•) Prediction : setosa [Highest Score 1.9]